SICK Safety Services

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Risk Assessment

Mechanical and electrical hazards in the conceptual machine design

Valeo

Customer:	**Valeo Friction Materials India Private Limited** No. 16A, Sengundram Industrial Area, Melrosapuram (via Singaperumal Koil), Chengalpattu / Kancheepuram District, 603204 Chennai Tamil Nadu India
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VERIFIED SAFETY
SICK 2

Table of contents

1 Project information

1.1 Contacts

Customer contact information		
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1.2 Version control

Version	Date	Description	Editor
0.1	8/12/2025	First Draft	Mr Muthu Vignesh R
0.2	8/12/2025	First Review	Mr.Jayesh Jani
0.3	8/14/2025	Second Review	Mr.Jayesh Jani

2 Introduction

2.1 Scope of service

(1) The risk assessment includes an analysis of the **mechanical hazards** and risks emanating from the respective machine, system or incomplete machine, as well as the **electrical hazards** and risks caused by the electrical equipment of the machine.
(2) The risk assessment is carried out on the basis of ISO 12100.
(3) The risk assessment is created on the basis of the explanations, descriptions and information as well as technical documents for the machine provided by the machine manufacturer. They are the essential elements for obtaining information and defining the minimum requirements for risk reduction measures for each machine. The overall effectiveness of the solutions recommended in this report is largely based on correct, accurate and complete information from the machine manufacturer.
(4) The assessment extends to the tasks on the machine specified by the machine manufacturer and coordinated with regard to the scope of services in the phases of setup, operation, cleaning, maintenance and troubleshooting of the machine life cycle phases in accordance with ISO 12100.
(5) SICK recommends measures to reduce risk in accordance with the 3-step method of ISO 12100 in order to create the conditions for the safe operation of a machine. Recommendations for risk-reducing measures are based on the currently relevant legal requirements, guidelines, requirements from standards and company-specific requirements (if applicable and known to SICK).
(6) For risk-reducing measures in the area of functional safety, a corresponding required Performance Level (PLr according to ISO 13849-1) is assigned on the basis of the risk identified.
(7) After a 4-eye check, the results are made available to the machine manufacturer in printed or electronic form (pdf) in the form of a report.
(8) Other hazard types that SICK cannot conclusively assess (e.g. hazards from heat, noise, hazardous substances, radiation, etc.) are identified and documented if they are obvious to SICK. SICK does not evaluate or recommend any measures to reduce the risk of these hazards. SICK recommends consulting a recognized specialist for further evaluation and protection.
(9) SICK does not provide any information on the static dimensioning and stability of the machine and the components contained in the machine.
(10) The risk assessment process aims to determine the level of an existing risk in order to make informed decisions about risk reduction. In some cases, a residual risk cannot be avoided, so that additional organizational measures are required (e.g. work instructions, signs, use of personal protective equipment, etc.).
(11) The risk reduction measures recommended by SICK are based on experience and the interpretation of the applicable standards. The aim of this report is to provide an overview of the health-endangering risks of a machine in order to develop a suitable strategy for protecting people who work on the machine. With the information in the report, the machine manufacturer can select suitable risk reduction measures and at the same time decide whether the subsequent residual risk of a machine is acceptable.
(12) Recommendations for risk reduction measures are always manufacturer-neutral.

Due to the restricted types of hazards taken into account, this risk assessment does **not represent a complete** risk assessment.

The following machines were evaluated as part of the risk assessment service. The evaluated risks and recommended risk reduction measures follow for each machine.

Machine type	Manufacturer	Machine name	Asset No.
Testing machine	ABB-Retrofit	QSW EM1 Test Bench Clutch test	Not Available

2.2 Impacts of legal requirements

In the present case, SICK has been commissioned to carry out a (partial) risk assessment on behalf of the client, based on the information provided by the client. All information provided by the client that has been incorporated into this risk assessment must be checked for correctness by the client.
The scope of the risk assessment carried out by SICK is limited to certain types of hazards and life cycle phases / tasks. It remains the responsibility of the machine manufacturer to evaluate all hazard types and activities not considered here in order to ensure a complete risk assessment.

2.3 Disclaimer

This risk assessment in the concept phase is intended to show the machine manufacturer dangers and risks in accordance with the agreed scope of services in order to assess which risk-reducing measures should be taken into account for the machine design.

(1) The results of this risk assessment are based exclusively on the information made available to us and the intended use notified to us. SICK assumes no liability for damage caused by incorrect, incomplete or inaccurate information or subsequent changes.

(2) The scope of services only takes mechanical and electrical hazards into account. Other types of hazard (such as noise, radiation, etc.) must additionally be considered and assessed by the machine manufacturer and are not the subject of this risk assessment.

(3) Identified risks are provided with appropriate risk-reducing measures by SICK. The machine manufacturer may have to take additional measures. The machine manufacturer must assess and decide whether the resulting residual risk is an acceptable / tolerable residual risk. Accordingly, SICK does not guarantee that the evaluated machine will be safe after the recommended risk reduction measures have been implemented. This assessment is ultimately the responsibility of the machine manufacturer and may require further considerations and / or measures.

(4) The machine manufacturer exempts SICK and its affiliated companies (as defined below) from all claims - including claims by third parties - that are based on non-compliance with the above information or on the inaccuracy or incompleteness of its information or on subsequent changes.

An "affiliated company" is a natural or legal person who controls another natural or legal person or is controlled by another natural or legal person or is jointly controlled by another natural or legal person with another natural or legal person. Control in this sense means the authority to influence the company and its business policy directly or indirectly on the basis of voting securities or voting rights, regardless of whether the authority was established by contract or otherwise.

3 Relevant Regulations and Standards

3.1 List of laws and regulations

Name	Date of Issue	Title
2006/42/EG	2006	Machinery Directive

3.2 List of standards

Name	Date of Issue	Title
EN ISO 12100	2010	Safety of machinery – General principles for design – Risk assessment and risk reduction
EN ISO 13849-1	2023	Safety of machinery - Safety-related parts of control systems - Part 1: General principles for design
EN 60204-1	2018	Safety of machinery – Electrical equipment of machines – Part 1: General requirements
EN ISO 13850	2015	Safety of machinery - Emergency stop function – Principles for design
EN ISO 13857	2019	Safety of machinery - Safety distances to prevent hazard zones being reached by upper and lower limbs
EN ISO 14118	2018	Safety of machinery - Prevention of unexpected start-up
EN ISO 14119	2013	Safety of machinery - Interlocking devices associated with guards - Principles for design and selection
EN ISO 14120	2015	Safety of machinery - Guards – General requirements for the design and construction of fixed and movable guards
EN ISO 14122-2	2016	Safety of machinery - Permanent means of access to machinery - Part 2: Working platforms and walkways
EN ISO 4414	2010	Pneumatic fluid power - General rules and safety requirements for systems and their components

4 Machine: QSW EM1 Test Bench Clutch test

4.1 Machine identification

<table>
<tr><th colspan="5">Machine data</th></tr>
<tr><td colspan="2">Machine name</td><td colspan="3">QSW EM1 Test Bench Clutch test</td></tr>
<tr><td colspan="2">Manufacturer</td><td colspan="3">ABB-Retrofit</td></tr>
<tr><td colspan="2">Machine type</td><td colspan="3">Testing machine</td></tr>
<tr><td colspan="2">Manufacturing date</td><td colspan="3">8/8/2025</td></tr>
<tr><td colspan="2">Re-manufacturing date</td><td colspan="3">8/8/2025</td></tr>
<tr><td colspan="2">Model No.</td><td colspan="3">Not Available</td></tr>
<tr><td colspan="2">Serial No.</td><td colspan="3">Not Available</td></tr>
<tr><td colspan="2">Asset No.</td><td colspan="3">Not Available</td></tr>
<tr><td rowspan="7">Asset location</td><td>Line</td><td colspan="3">Valeo Friction Materials India private Ltd. - R&D</td></tr>
<tr><td>Building</td><td colspan="3">R&D Cell</td></tr>
<tr><td>Factory</td><td colspan="3">No. 16A, Sengundram Industrial Area,</td></tr>
<tr><td>Street</td><td colspan="3">Melrosapuram (via Singaperumal Koil),
Chengalpattu / Kancheepuram District,</td></tr>
<tr><td>City</td><td>Chennai</td><td>Zip code</td><td>603204</td></tr>
<tr><td>State</td><td colspan="3">Tamil Nadu</td></tr>
<tr><td>Country</td><td colspan="3">India</td></tr>
</table>

4.2 Technical machine documentation

The risk assessment is based on the following documents provided by the machine manufacturer.

Documents	Version	Date

4.3 Limits of the Machine

<table>
<tr><th colspan="7">Limits of the Machine</th></tr>
<tr><td>Intended use</td><td colspan="6">To perform thermal and mechanical endurance tests on clutch assemblies using a controlled environment. It validates performance, wear, and ill-start behavior under predefined parameters such as temperature, RPM, and inertia settings in both manual and auto modes.</td></tr>
<tr><td>Reasonably foreseeable misuse</td><td colspan="6">Running the machine with door unlocked, bypassing interlocks, skipping temperature ramp-up, incorrect adapter installation, or switching to auto mode without completing startup checks, leading to unsafe test launches or false data recording.</td></tr>
<tr><td>Expected/ known malfunctions, errors & faults</td><td colspan="6">Door lock sensor failure, RTD sensor mismatch, motor overspeed, adapter loosening, bearing wear, or software startup lag (Morphee). Inconsistent temperature rise or faulty RPM readings may trigger test abort or unsafe shutdown.</td></tr>
<tr><td>Time limits</td><td>Days / year</td><td>200</td><td>Hours / day</td><td>24</td><td>Shifts / day</td><td>1</td></tr>
<tr><td>Space limits</td><td colspan="6">Machine Footprint: Approx. 3 m (L) × 2.5 m (W)

Operating Clearance: Add 1 m on all sides for safe access, maintenance, and airflow

Ideal Total Space: 5 m (L) × 4.5 m (W) = ~22.5 m²</td></tr>
<tr><td>Environmental conditions</td><td colspan="6">Ambient Temperature: 20°C to 35°C (stable room temperature recommended)

Humidity: 40% to 70% RH, non-condensing

Ventilation: Proper air circulation required; forced exhaust for high-temperature operations

Cleanliness: Dust-free environment; avoid corrosive gases or oil mist

Power Supply: Stable 3-phase power with proper earthing

Noise Control: Acoustic enclosure recommended if >80 dB during operation</td></tr>
</table>

Machine parameters		
Processed workpieces/ raw materials		Test bench for clutch of different variants/impregnated yarn made of glass fiber,infused with metal
Capacity		Monthly 3 or 4 tests with continuous intervals
Manufacturing process description		The test begins with a reset to energize the SSB electrical panel, switching the machine to the ON state. In manual mode, the operator opens the test chamber, places the test part cover, installs the clutch assembly and friction plate, verifies parameters, and sets the required inertia value. The door is locked, and mode selection is confirmed before switching to auto mode for the Morphy software start. A startup test is run to reach the target temperature (20°C–500°C) using four RTDs and K-type sensors, and the motor speed (100–300 RPM) is verified. After startup validation, the main test begins. Three types of tests are performed — 1. Performance Test, 2. Ill Start Test, 3. Wear Test Each following the same setup. Test durations range from minimum 1 day to 15 days, with parameter checks happening once in every two days in manual mode. Routine maintenance includes monthly adapter fixture changes, adapter bearing replacement every 10–15 days, and consistent greasing settings across variants.
Operating modes		Manual Automatic
Power supply	**Electrical**	415 V AC
	Pneumatic	6 Bar
	Hydraulic	Not Available
	Other	Not Available

4.4 Picture(s)

Picture(s)	
EM1 Test Bench	
	
Test Bench in Guards open stage	Test bench has 3 doors interlocked with solenoid interlocking type
	
Test component loading zone	Components to be tested are loaded here with the operator control panel

NA	Retrofit machine from other Valeo Location so no Name plate

4.5 Tasks performed at the machine

Affected persons	Skill	Skill description
Operator 1	Instructed	Trained staff in test bench with work instructions manual Bi - yearly Technical background with good engineering practices
Maintenance person	Skilled	Trained staff in electrical maintenance and good engineering practices Bi - yearly with Technical background.
House keeping staff	Unskilled	Cleaning staff floating man power Non technical person
Third party Vendor-Calibration	Skilled	Technical staff accredited by NABL certified body, for calibration of torque pressure temperature and other parameter instruments
Operator 2	Instructed	Expert training qualified for critical parameters modification in HMI
Visitors	Unskilled	Visitors not trained to access the machine parts or door

Icon	Task	Affected persons
	Task group - Task- Air filter cleaning The air filter is replaced periodically to maintain airflow quality and prevent dust accumulation inside the test machine. The operator who is trained for this activity stops the machine, isolates it from power/air supply, and opens the filter housing access panel. The used filter is removed and disposed of as per waste handling guidelines, and a new filter is inserted in the correct orientation. The housing is secured, and power/air is restored. Final checks are done to ensure proper sealing and airflow before resuming operation. **Dedicated tasks to this group** • Cleaning/Maintenance	
	Component Load/ Unload process Before the start of the continuous testing test piece / clutch component is placed in the machine adapter area in manual mode series of steps are performed by skilled operator who understands the hazards involved	Operator 2 Operator 1
	Observation Process monitoring / observation. - Overview of test process if the filter is functioning properly since it is a continuous process once in the start of the day or end of the day Calibration of temperature or field sensors	Operator 1 Third party Vendor-Calibration Maintenance person Operator 2
	Trouble shoot Fault-finding / Troubleshooting once if the fault occurs in the system mostly third party is preferred for major trouble shoot in an interval of once in 6 months	Maintenance person
	Cleaning/Maintenance Cleaning / Maintenance - cleans the left over dusts in the machine perimeter as well the surface area of the machine is cleaned dusting is done	House keeping staff Operator 1
	Setting- Test part - Startup test Setting, Teaching / programming and / or process changeover. either 24 hrs once for few tests and 15 days once for some tests the startup test is being performed before starting the actual test case scenario total setup time taken more than one hour.	Operator 2

	Operator 2 requires to open the door in manual mode for activation of stroke lever holding test parts while the doors is opened movement of cylinder and fixed part can introduce crush injury in the machine.	
	**Presence at the Machine** Persons being present in a hazardous area at or near the machine without a dedicated task at the machine / not working at the machine. visitors or adjacent team members for parameter collection or delivery of test component	Visitors Third party Vendor-Calibration

5 Risk analysis

5.1 Hazard list according to ISO 12100

Hazard origin	Risk ID	Hazard source	Task	Risk IN	Risk OUT
Mechanical hazards					
Approach of a moving element to a fixed part	**R017**	Pneumatic cylinder	Component Load/ Unload process	**5**	**<1**
	R018	Pneumatic cylinder	Setting- Test part - Startup test	**5**	**<1**
Kinetic energy - Intertia present in the motors no brakes present	**R001**	Rotating parts	Component Load/ Unload process	**5**	**<1**
	R002	Rotating parts	Trouble shoot	**4**	**<1**
	R019	Rotating parts	Cleaning/Maintenance	**4**	**<1**
	R020	Rotating parts	Setting- Test part - Startup test	**5**	**<1**
Moving elements- power transfer shaft	**R003**	Transfer shaft in motion	Component Load/ Unload process	**5**	**<1**
	R004	Transfer shaft in motion	Trouble shoot	**4**	**<1**
	R021	Transfer shaft in motion	Cleaning/Maintenance	**4**	**<1**
Electrical hazards					
Live parts- in control panel	**R005**	3 phase Power supply	Trouble shoot	**7**	**<1**
Live parts under high voltage	**R006**	Over voltage from Grid / power fluctuation	Trouble shoot	**4**	**<1**
Thermal radiation- Heat due to friction movement internal	**R008**	Internal friction induced heat - 500 degrees	Component Load/ Unload process	**5**	**<1**
	R009	Internal friction induced heat - 500 degrees	Trouble shoot	**4**	**<1**
	R010	Internal friction induced heat - 500 degrees	Cleaning/Maintenance	**4**	**<1**
Noise hazards					
Manufacturing process	**R013**	Main Drive under high RPM	Presence at the Machine	**1**	**<1**
Material/substance hazards					
Dust	**R015**	From the testing process	Observation	**1**	**<1**
	R016	From the testing process	Presence at the Machine	**1**	**<1**
	R022	From the testing process	Cleaning/Maintenance	**1**	**<1**

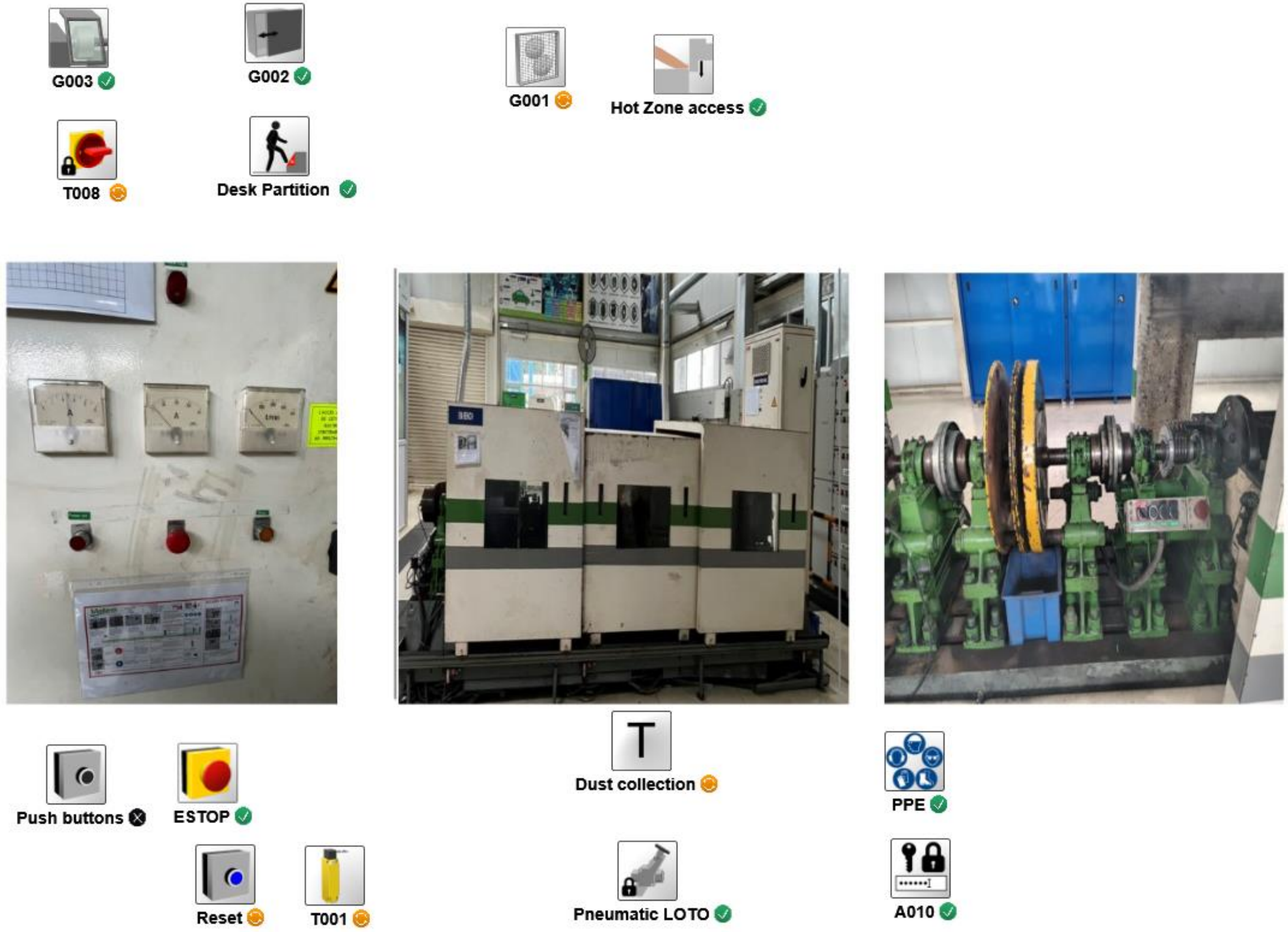
Risk level	Risk level definition
<1	Very low
1	Low
2-3	Increased
4-7	High
8-10	Very high

The estimated residual risk (Risk OUT) is not to be equated with an acceptable residual risk. This applies to all risk levels, even 0.
Due to the subjective determination of whether an acceptable risk has been reached, SICK cannot determine whether the achievable residual risk is an acceptable risk for the machine manufacturer. As a partner with specialized experience and technical expertise, SICK can provide assistance and recommendations. Ultimately, however, accepting (or rejecting) the residual risk is the sole responsibility of the machine manufacturer.

5.2 Overview of identified hazards and tasks

Tasks & Hazards


RRM - Machine Overview



5.3 Identified risks

5.3.1 R001

Mechanical hazards	
Kinetic energy - Intertia present in the motors no brakes present	
Hazard source	**Consequence**
Rotating parts	Crushing
Main motor/drive rotation , power transfer shaft, test part rotating end these movements does not have break potential to stop right after the stop activation of the machine it takes few seconds to stop the rotating movements	
Component Load/ Unload process	
Before the start of the continuous testing test piece / clutch component is placed in the machine adapter area in manual mode series of steps are performed by skilled operator who understands the hazards involved	**Affected persons** Operator 2 Operator 1

Severity [S]	Exposure [E]	Avoidance [A]	Occurrence [O]	Risk IN
S3	E2	A2	O3	5

Risk reduction					S	E	A	O	Risk OUT
Inherent design					S3	E0	A1	O1	<1
Engineering	Guards	G002	Movable guard sliding						
		G001	Fixed guard						
	Devices	T001	Interlock - Guard locking						
		T004	E-Stop push button						
		Reset	Reset button						
		Push buttons	Hold-to-run control						
Administrative		Security screws	Tamper resistant fastener						
		PPE	PPE - General						
		A006	Safe work procedures						
		A001	Instruction manual						
		Crush Injury	Awareness (safety) signs						
Residual risk	if the interlocks are bypassed with the tamper proof screw tool the doors can be accessed before the stop time is reached								

5.3.2 R002

Mechanical hazards	
Kinetic energy - Intertia present in the motors no brakes present	
Hazard source	**Consequence**
Rotating parts	Crushing
Main motor/drive rotation , power transfer shaft, test part rotating end these movements does not have break potential to stop right after the stop activation of the machine it takes few seconds to stop the rotating movements	
Trouble shoot	
Fault-finding / Troubleshooting once if the fault occurs in the system mostly third party is preferred for major trouble shoot in an interval of once in 6 months	**Affected persons** Maintenance person

Severity [S]	Exposure [E]	Avoidance [A]	Occurrence [O]	Risk IN
S3	E1	A2	O3	4

Risk reduction				S	E	A	O	Risk OUT
Inherent design								
Engineering	**Guards**	G001	Fixed guard	S3	E0	A1	O3	<1
		G003	Shield					
		G002	Movable guard sliding					
	Devices	T001	Interlock - Guard locking					
		Reset	Reset button					
		ESTOP	E-Stop push button					
Administrative		A001	Instruction manual					
		A006	Safe work procedures					
		Guards	Awareness (safety) signs					
		Security screws	Tamper resistant fastener					
Residual risk	If the interlocks are bypassed with the tamper proof screw tool the doors can be accessed before the stop time is reached							

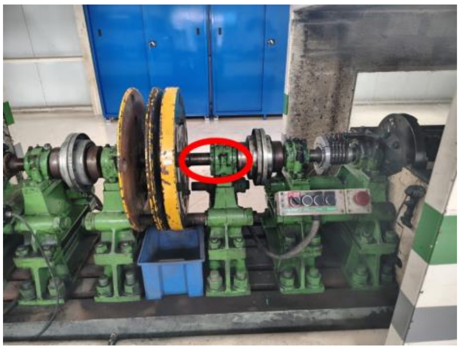
5.3.3 R003

Mechanical hazards		
Moving elements- power transfer shaft		
Hazard source	**Consequence**	
Transfer shaft in motion	Drawing-in or trapping	
Reach to the transfer shaft can cause drawing or trapping hazard during maintenance or adapter change process		
Component Load/ Unload process		
Before the start of the continuous testing test piece / clutch component is placed in the machine adapter area in manual mode series of steps are performed by skilled operator who understands the hazards involved	**Affected persons** Operator 2 Operator 1	

Severity [S]	Exposure [E]	Avoidance [A]	Occurrence [O]	Risk IN
S3	E2	A2	O3	5

Risk reduction				S	E	A	O	Risk OUT
Inherent design		Hot Zone access	Prevention through design	S3	E0	A1	O1	<1
Engineering	**Guards**	G001	Fixed guard					
		G002	Movable guard sliding					
		G003	Shield					
	Devices	T001	Interlock - Guard locking					
		Push buttons	Hold-to-run control					
		ESTOP	E-Stop push button					
		Reset	Reset button					
Administrative		A006	Safe work procedures					
		A001	Instruction manual					
		Guards	Awareness (safety) signs					
		Crush Injury	Awareness (safety) signs					
Residual risk	if the fixed guards are not in place possible reach to moving parts							

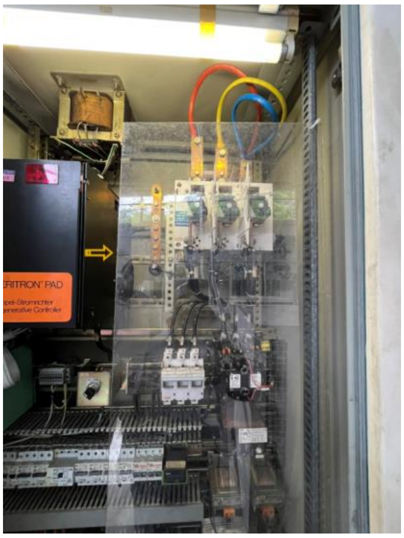
5.3.4 R004

Mechanical hazards		
Moving elements- power transfer shaft		
Hazard source	**Consequence**	
Transfer shaft in motion	Drawing-in or trapping	
Reach to the transfer shaft can cause drawing or trapping hazard during maintenance or adapter change process		
Trouble shoot		
Fault-finding / Troubleshooting once if the fault occurs in the system mostly third party is preferred for major trouble shoot in an interval of once in 6 months	**Affected persons** Maintenance person	

Severity [S]	Exposure [E]	Avoidance [A]	Occurrence [O]	Risk IN
S3	E1	A2	O3	4

Risk reduction				S	E	A	O	Risk OUT
Inherent design								
Engineering	**Guards**	G001	Fixed guard					
		G003	Shield					
		G002	Movable guard sliding					
	Devices	T001	Interlock - Guard locking					
		Pneumatic LOTO	Pneumatic shut-off and exhaust valve					
		Reset	Reset button					
		ESTOP	E-Stop push button	S3	E0	A1	O1	<1
Administrative		A006	Safe work procedures					
		A001	Instruction manual					
		Guards	Awareness (safety) signs					
		Crush Injury	Awareness (safety) signs					
		PPE	PPE - General					
		A010	Measures to avoid unauthorized access					
		LOTO Policy	Lockout / Tagout					
Residual risk	if the fixed guards are not in place possible reach to moving parts							

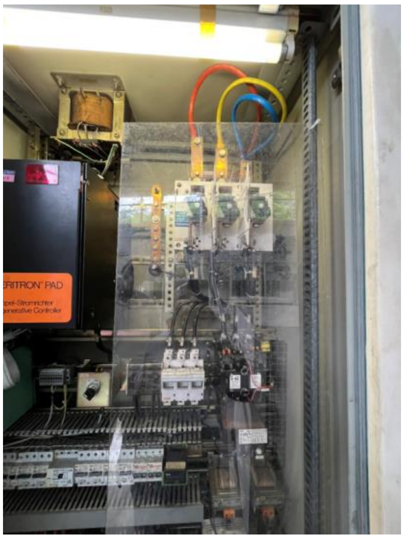
5.3.5 R005

Electrical hazards		
Live parts- in control panel		
Hazard source	**Consequence**	
3 phase Power supply	Electrocution	
Possible reach to the live terminals of bus bar RYB basic protection is present but possible to reach the bus bar no IP 2X control is not present		
Trouble shoot		
Fault-finding / Troubleshooting once if the fault occurs in the system mostly third party is preferred for major trouble shoot in an interval of once in 6 months	**Affected persons** Maintenance person	

Severity [S]	**Exposure [E]**	**Avoidance [A]**	**Occurrence [O]**	**Risk IN**
S4	E1	A2	O3	7

Risk reduction				S	E	A	O	Risk OUT
Inherent design				S4	E0	A1	O1	<1
Engineering	**Guards**	G004-Busbar acc	Fixed guard					
	Devices	T008	Emergency off					
Administrative		A005	Lockout / Tagout					
		A006	Safe work procedures					
		A001	Instruction manual					
		PPE	PPE - General					
Residual risk	if the fixed acrylic guards are not in place and earthing is missed electric shock is possible							

5.3.6 R006

Electrical hazards		
Live parts under high voltage		
Hazard source	**Consequence**	
Over voltage from Grid / power fluctuation	Shock	
Power fluctuation controls are missing in the system which can potentially damage the control systems and the respective actuators		
Trouble shoot		
Fault-finding / Troubleshooting once if the fault occurs in the system mostly third party is preferred for major trouble shoot in an interval of once in 6 months	**Affected persons** Maintenance person	

Severity [S]	Exposure [E]	Avoidance [A]	Occurrence [O]	Risk IN
S3	E1	A2	O3	4

Risk reduction				S	E	A	O	Risk OUT
Inherent design				S3	E0	A1	O1	<1
Engineering	**Guards**	G004-Busbar acc	Fixed guard					
	Devices	on board LOTO	Emergency off					
Administrative		A006	Safe work procedures					
		A001	Instruction manual					
		A005	Lockout / Tagout					
Residual risk	if the fixed acrylic guards are not in place and earthing is missed electric shock is possible							

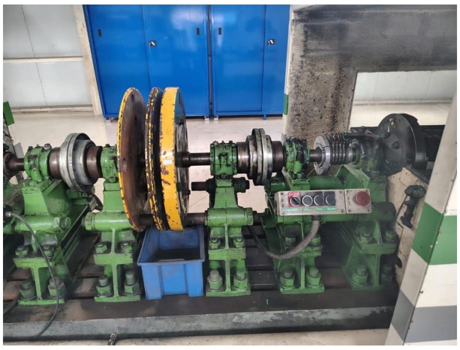
5.3.7 R008

Electrical hazards

Thermal radiation- Heat due to friction movement internal

Hazard source	Consequence
Internal friction induced heat - 500 degrees	Burn

During the testing process multiple rotation drives induce frictional heat in the system which is unaccessible but potentially beyond the tolerable temperatures of human being - until the temperature reduces for a tolerable region door cant be opened with help of guard locking units - override of this function is only with dept user by bringing manual mode door monitoring is missing but the logic ensures that the machine will only function when all the door interlocks are healthy and closed



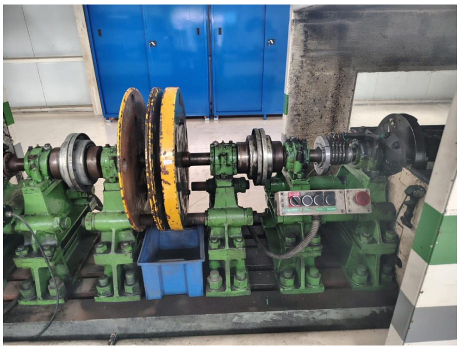
Component Load/ Unload process

Description	Affected persons
Before the start of the continuous testing test piece / clutch component is placed in the machine adapter area in manual mode series of steps are performed by skilled operator who understands the hazards involved	Operator 2 Operator 1

Severity [S]	Exposure [E]	Avoidance [A]	Occurrence [O]	Risk IN
S3	E2	A2	O3	5

Risk reduction				S	E	A	O	Risk OUT
Inherent design		Hot Zone access	Prevention through design	S3	E0	A1	O1	<1
Engineering	Guards	G003	Shield					
Engineering	Guards	G002	Movable guard sliding					
Engineering	Devices	T001	Interlock - Guard locking					
Administrative		A006	Safe work procedures					
Administrative		A001	Instruction manual					
Administrative		USER PW	Measures to avoid unauthorized access					
Administrative		LOTO Policy	Lockout / Tagout					
Residual risk		Low energy task; main hazards are dust if PPE are not worn properly						

5.3.8 R009

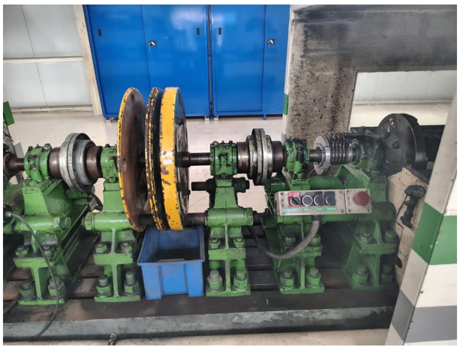
Electrical hazards		
Thermal radiation- Heat due to friction movement internal		
Hazard source	**Consequence**	
Internal friction induced heat - 500 degrees	Burn	
During the testing process multiple rotation drives induce frictional heat in the system which is unaccessible but potentially beyond the tolerable temperatures of human being - until the temperature reduces for a tolerable region door cant be opened with help of guard locking units - override of this function is only with dept user by bringing manual mode door monitoring is missing but the logic ensures that the machine will only function when all the door interlocks are healthy and closed		
Trouble shoot		
Fault-finding / Troubleshooting once if the fault occurs in the system mostly third party is preferred for major trouble shoot in an interval of once in 6 months	**Affected persons** Maintenance person	

Severity [S]	Exposure [E]	Avoidance [A]	Occurrence [O]	Risk IN
S3	E1	A2	O3	4

Risk reduction				S	E	A	O	Risk OUT
Inherent design		Hot Zone access	Prevention through design	S3	E0	A1	O1	<1
Engineering	**Guards**	G002	Movable guard sliding					
Engineering	**Devices**	T001	Interlock - Guard locking					
Administrative		A005	Lockout / Tagout					
Administrative		A006	Safe work procedures					
Administrative		A001	Instruction manual					
Administrative		Guards	Awareness (safety) signs					
Administrative		Hot surface	Awareness (safety) signs					
Residual risk	if the fixed guards are not put in place possible reach to heat surfaces							

5.3.9 R010

Electrical hazards	
Thermal radiation- Heat due to friction movement internal	
Hazard source	**Consequence**
Internal friction induced heat - 500 degrees	Burn
During the testing process multiple rotation drives induce frictional heat in the system which is unaccessible but potentially beyond the tolerable temperatures of human being - until the temperature reduces for a tolerable region door cant be opened with help of guard locking units - override of this function is only with dept user by bringing manual mode door monitoring is missing but the logic ensures that the machine will only function when all the door interlocks are healthy and closed	
Cleaning/Maintenance	
Cleaning / Maintenance - cleans the left over dusts in the machine perimeter as well the surface area of the machine is cleaned dusting is done	**Affected persons** House keeping staff Operator 1



Severity [S]	Exposure [E]	Avoidance [A]	Occurrence [O]	Risk IN
S3	E1	A2	O3	4

Risk reduction				S	E	A	O	Risk OUT
Inherent design		Hot Zone access	Prevention through design	S3	E0	A1	O1	<1
Engineering	Guards	G002	Movable guard sliding					
	Devices	T001	Interlock - Guard locking					
Administrative		A006	Safe work procedures					
		A001	Instruction manual					
		Hot surface	Awareness (safety) signs					
		PPE	PPE - General					
Residual risk	if the fixed guards are not put in place possible reach to hot surfaces							

5.3.10 R013

Noise hazards		
Manufacturing process		
Hazard source	**Consequence**	
Main Drive under high RPM	Discomfort	
During the process of testing at high speeds of 3000 RPM there is a high potential of noise hazard suitable personal protective equipment's are to be used		
Presence at the Machine		
Persons being present in a hazardous area at or near the machine without a dedicated task at the machine / not working at the machine. visitors or adjacent team members for parameter collection or delivery of test component	**Affected persons** Visitors Third party Vendor-Calibration	

Severity [S]	Exposure [E]	Avoidance [A]	Occurrence [O]	Risk IN
S2	E1	A2	O3	1

Risk reduction				S	E	A	O	Risk OUT
Inherent design		Desk Partition	Elimination / Limiting Interaction	S2	E1	A2	O3	<1
Engineering	Guards	G003	Shield					
		G002	Movable guard sliding					
	Devices							
Administrative		PPE	PPE - General					
		Guards	Awareness (safety) signs					
		Crush Injury	Awareness (safety) signs					
		Hot surface	Awareness (safety) signs					
Residual risk		if the fixed guards are not in place possible reach to moving parts						

5.3.11 R015

Material/substance hazards		
Dust		
Hazard source	**Consequence**	
From the testing process	Breathing difficulties, suffocation	
During the test process fine dust particles are present in the area suitable dust collection systems are present in the machine - fixed pipeline to be modified and inspection to be done at regular intervals		
Observation		
Process monitoring / observation. - Overview of test process if the filter is functioning properly since it is a continuous process once in the start of the day or end of the day Calibration of temperature or field sensors	**Affected persons** Operator 1 Third party Vendor-Calibration Maintenance person Operator 2	

Severity [S]	Exposure [E]	Avoidance [A]	Occurrence [O]	Risk IN
S2	E1	A2	O3	1

Risk reduction				S	E	A	O	Risk OUT
Inherent design		Desk Partition	Elimination / Limiting Interaction	S2	E1	A2	O3	<1
Engineering	**Guards**	G003	Shield					
		G002	Movable guard sliding					
	Devices	Dust collection	Exhaust system					
Administrative		PPE	PPE - General					
		I&M	Instruction manual					
		A006	Safe work procedures					
		A001	Instruction manual					
Residual risk		Low energy task; main hazards are dust if PPE are not worn properly						

5.3.12 R016

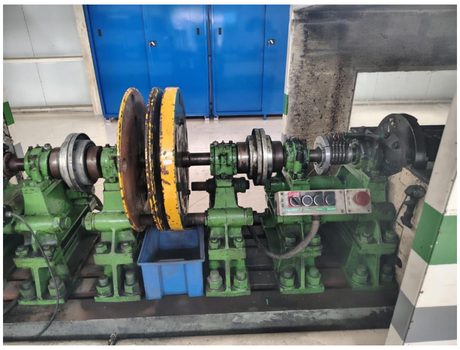
Material/substance hazards	
Dust	
Hazard source	**Consequence**
From the testing process	Breathing difficulties, suffocation
During the test process fine dust particles are present in the area suitable dust collection systems are present in the machine - fixed pipeline to be modified and inspection to be done at regular intervals	
Presence at the Machine	
Persons being present in a hazardous area at or near the machine without a dedicated task at the machine / not working at the machine. visitors or adjacent team members for parameter collection or delivery of test component	**Affected persons** Visitors Third party Vendor-Calibration

Severity [S]	Exposure [E]	Avoidance [A]	Occurrence [O]	Risk IN
S2	E1	A2	O3	1

Risk reduction				S	E	A	O	Risk OUT
Inherent design				S2	E1	A2	O3	<1
Engineering	Guards	G003	Shield					
		G002	Movable guard sliding					
	Devices	Dust collection	Exhaust system					
Administrative		PPE	PPE - General					
		A006	Safe work procedures					
		A001	Instruction manual					
Residual risk	Low energy task; main hazards are dust if PPE are not worn properly							

5.3.13 R017

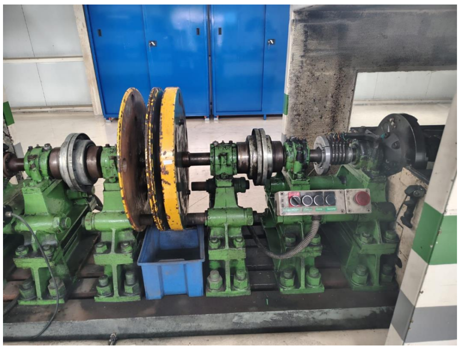
Mechanical hazards		
Approach of a moving element to a fixed part		
Hazard source	**Consequence**	
Pneumatic cylinder	Crushing	
During loading of test component possible reach of fingers or hand between the moving cylinder and fixed adapter holding area		
Component Load/ Unload process		
Before the start of the continuous testing test piece / clutch component is placed in the machine adapter area in manual mode series of steps are performed by skilled operator who understands the hazards involved	**Affected persons** Operator 2 Operator 1	



Severity [S]	Exposure [E]	Avoidance [A]	Occurrence [O]	Risk IN
S3	E2	A2	O3	5

Risk reduction				S	E	A	O	Risk OUT
Inherent design								
Engineering	Guards	G002	Movable guard sliding	S3	E0	A1	O1	<1
		G003	Shield					
	Devices	T001	Interlock - Guard locking					
		Push buttons	Hold-to-run control					
		ESTOP	E-Stop push button					
		Reset	Reset button					
Administrative		A006	Safe work procedures					
		A001	Instruction manual					
		Security screws	Tamper resistant fastener					
		PPE	PPE - General					
		Crush Injury	Awareness (safety) signs					
		A010	Measures to avoid unauthorized access					
Residual risk	if the push button control is manipulated or safe work procedures are not followed							

5.3.14 R018

Mechanical hazards		
Approach of a moving element to a fixed part		
Hazard source	**Consequence**	
Pneumatic cylinder	Crushing	
During loading of test component possible reach of fingers or hand between the moving cylinder and fixed adapter holding area		
Setting- Test part - Startup test		
Setting, Teaching / programming and / or process changeover. either 24 hrs once for few tests and 15 days once for some tests the startup test is being performed before starting the actual test case scenario total setup time taken more than one hour. Operator 2 requires to open the door in manual mode for activation of stroke lever holding test parts while the doors is opened movement of cylinder and fixed part can introduce crush injury in the machine.	**Affected persons** Operator 2	

Severity [S]	Exposure [E]	Avoidance [A]	Occurrence [O]	Risk IN
S3	E2	A2	O3	5

Risk reduction					S	E	A	O	Risk OUT
Inherent design									
Engineering	**Guards**	G003	Shield		S3	E0	A1	O1	<1
		G002	Movable guard sliding						
	Devices	Push buttons on board	Hold-to-run control						
		LOTO	Emergency off						
		ESTOP	E-Stop push button						
		Reset	Reset button						
		T001	Interlock - Guard locking						

Administrative	A006	Safe work procedures					
	A001	Instruction manual					
	USER PW	Measures to avoid unauthorized access					
	I&M	Instruction manual					
	Security screws	Tamper resistant fastener					
	A010	Measures to avoid unauthorized access					
	PPE	PPE - General					
	Crush Injury	Awareness (safety) signs					
Residual risk	if the push button control is manipulated or safe work procedures are not followed						

5.3.15 R019

Mechanical hazards		
Kinetic energy - Intertia present in the motors no brakes present		
Hazard source	**Consequence**	
Rotating parts	Crushing	
Main motor/drive rotation , power transfer shaft, test part rotating end these movements does not have break potential to stop right after the stop activation of the machine it takes few seconds to stop the rotating movements		
Cleaning/Maintenance		
Cleaning / Maintenance - cleans the left over dusts in the machine perimeter as well the surface area of the machine is cleaned dusting is done	**Affected persons** House keeping staff Operator 1	

Severity [S]	Exposure [E]	Avoidance [A]	Occurrence [O]	Risk IN
S3	E1	A2	O3	4

Risk reduction				S	E	A	O	Risk OUT
Inherent design								
Engineering	Guards	G003	Shield	S3	E0	A1	O2	<1
		G002	Movable guard sliding					
	Devices	T001	Interlock - Guard locking					
		Reset	Reset button					
		ESTOP	E-Stop push button					
		T008	Emergency off					
Administrative		Security screws	Tamper resistant fastener					
		LOTO Policy	Lockout / Tagout					
		A010	Measures to avoid unauthorized access					
		A006	Safe work procedures					
		A001	Instruction manual					
		Guards	Awareness (safety) signs					
		PPE	PPE - General					
Residual risk	Minor injury possible if the guards are not in place or procedures are not followed							

5.3.16 R020

Mechanical hazards

Kinetic energy - Intertia present in the motors no brakes present

Hazard source	Consequence
Rotating parts	Crushing

Main motor/drive rotation , power transfer shaft, test part rotating end these movements does not have break potential to stop right after the stop activation of the machine it takes few seconds to stop the rotating movements

Setting- Test part - Startup test

	Affected persons
Setting, Teaching / programming and / or process changeover. either 24 hrs once for few tests and 15 days once for some tests the startup test is being performed before starting the actual test case scenario total setup time taken more than one hour. Operator 2 requires to open the door in manual mode for activation of stroke lever holding test parts while the doors is opened movement of cylinder and fixed part can introduce crush injury in the machine.	Operator 2

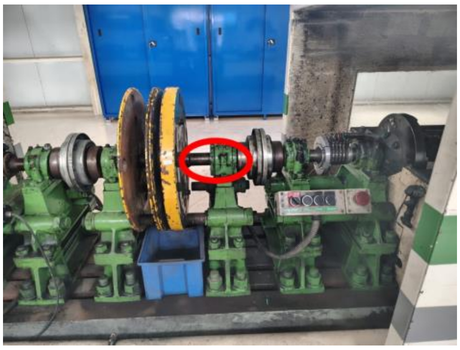
Severity [S]	Exposure [E]	Avoidance [A]	Occurrence [O]	Risk IN
S3	E2	A2	O3	5

Risk reduction					S	E	A	O	Risk OUT
Inherent design									
Engineering	Guards	G002	Movable guard sliding		S3	E0	A2	O1	<1
		G003	Shield						
		G001	Fixed guard						
	Devices	Push buttons	Hold-to-run control						
		ESTOP	E-Stop push button						
		Reset	Reset button						
		T001	Interlock - Guard locking						

Administrative	Security screws	Tamper resistant fastener						
	PPE	PPE - General						
	LOTO Policy	Lockout / Tagout						
	Guards	Awareness (safety) signs						
	A006	Safe work procedures						
	A001	Instruction manual						
	A010	Measures to avoid unauthorized access						
Residual risk	Minor injury possible if the guards are not in place or procedures are not followed							

5.3.17 R021

Mechanical hazards		
Moving elements- power transfer shaft		
Hazard source	**Consequence**	
Transfer shaft in motion	Drawing-in or trapping	
Reach to the transfer shaft can cause drawing or trapping hazard during maintenance or adapter change process		
Cleaning/Maintenance		
Cleaning / Maintenance - cleans the left over dusts in the machine perimeter as well the surface area of the machine is cleaned dusting is done	**Affected persons**	
	House keeping staff Operator 1	



Severity [S]	Exposure [E]	Avoidance [A]	Occurrence [O]	Risk IN
S3	E1	A2	O3	4

Risk reduction					S	E	A	O	Risk OUT
Inherent design									
Engineering	Guards	G002	Movable guard sliding						
		G003	Shield						
	Devices	ESTOP	E-Stop push button						
		Reset	Reset button						
		T001	Interlock - Guard locking						
		Pneumatic LOTO	Pneumatic shut-off and exhaust valve						
		T008	Emergency off		S3	E0	A1	O1	<1
Administrative		PPE	PPE - General						
		LOTO Policy	Lockout / Tagout						
		A006	Safe work procedures						
		A001	Instruction manual						
		A010	Measures to avoid unauthorized access						
		Guards	Awareness (safety) signs						
		Security screws	Tamper resistant fastener						
Residual risk	Minor injury possible if the guards are not in place or procedures are not followed								

5.3.18 R022

Material/substance hazards		
Dust		
Hazard source	**Consequence**	
From the testing process	Breathing difficulties, suffocation	
During the test process fine dust particles are present in the area suitable dust collection systems are present in the machine - fixed pipeline to be modified and inspection to be done at regular intervals		
Cleaning/Maintenance		
Cleaning / Maintenance - cleans the left over dusts in the machine perimeter as well the surface area of the machine is cleaned dusting is done	**Affected persons** House keeping staff Operator 1	

Severity [S]	Exposure [E]	Avoidance [A]	Occurrence [O]	Risk IN
S2	E1	A2	O3	1

Risk reduction				S	E	A	O	Risk OUT
Inherent design				S2	E1	A2	O3	<1
Engineering	**Guards**	G001	Fixed guard					
		G003	Shield					
		G002	Movable guard sliding					
	Devices							
Administrative		PPE	PPE - General					
		A001	Instruction manual					
		A006	Safe work procedures					
		Crush Injury	Awareness (safety) signs					
		Guards	Awareness (safety) signs					
Residual risk	if Right PPE are not worn and dust collection systems are not maintained, exposed to the area for long duration possible injury in the respiratory systems							

6 Risk reduction

This chapter describes the risk reduction measures referenced in the individual risk tables. The overviews show the general locations of the measures at the machine.

6.1 Overview of risk reduction measures

Machine Overview

6.2 Risk reduction measures

	ID	Measure
Inherent safe design		
	Desk Partition	**Elimination / Limiting Interaction** Door partition is designed and provided to limit the operation of the machine from the machine perimeter area where noise is higher operator operates and observes the machine parameters from the control desk Typical residual risk: - Modifications to the workplace that allow standing/climbing
	Hot Zone access	**Prevention through design** Accessible parts of machinery shall have no sharp edges, no sharp angles, no rough surfaces, no protruding parts likely to cause injury, and no openings which can "trap" parts of the body or clothing. Crushing and shearing hazards to be avoided via appropriate geometrical design, by proper design of the form and relative location of the mechanical moving components. Typical residual risk: - Modification of the geometrical design
Technical protective measures / Engineering controls – Guards		
G	Filter guarding	**Guarding generic** A generic guarding additional to the air filter to be added which will prevent excess dust during the filter change
	G001	**Fixed guard** Fixed guards are in place to prevent access to the moving parts with a lever but chances are there that lever could be removed
	G002	**Movable guard sliding** 3 Movable and sliding doors are present with mechanical stopper and interlocks in place openings are present but it is designed as per EN ISO 13855 **Assigned Safety Function(s):** SF10 – Initiating a stop
	G003	**Shield** installed designed and positioned according to ISO 14120 and ISO 13857. For existing measures: Typical residual risk: - Failure to replace guard following maintenance, repair or setup - Failure to adjust guard before operation - Insufficient inspection and maintenance - Failure due to deterioration of material
	G004-Busbar acc	**Fixed guard** Acrylic To be designed and positioned according to ISO 14120 and ISO 13857 across live terminals For existing measures: Height: xxx mm Length: xxx mm Mesh gap dimensions: xx mm Make of the guard: xx Typical residual risk: - Failure to replace guard following maintenance, repair or setup
	G004-Busbar acc	**Fixed guard** Acrylic To be designed and positioned according to ISO 14120 and ISO 13857 across live terminals For existing measures: Height: xxx mm Length: xxx mm Mesh gap dimensions: xx mm Make of the guard: xx Typical residual risk: - Failure to replace guard following maintenance, repair or setup

Technical protective measures / Engineering controls – Devices			
T	Dust collection	**Exhaust system** Dust collection system present in the machine to exhaust dust from the room	-none-
	ESTOP	**E-Stop push button** Additional estop button to be implemented according to ISO 13850 and IEC 60204-1. Typical residual risk: - Operation for purposes other than emergency - Device not readily accessible - Introduction of obstruction preventing actuation **Assigned Safety Function(s):** SF99 – Emergency stop	PLr d
	on board LOTO	**Emergency off** To be implemented according to IEC 60204-1. Typical residual risk: - Failure to follow documented procedures - Operation for purposes other than emergency - Device not readily accessible	
T	Over voltage PR	**Device generic** An over votlage protective surge device measure to be used if no suitable RRM is found in the machine for high voltage risk.	-none-
T	Over voltage PR	**Device generic** An over votlage protective surge device measure to be used if no suitable RRM is found in the machine for high voltage risk.	-none-
	Pneumatic LOTO	**Pneumatic shut-off and exhaust valve** Primary machine pneumatic energy disconnection device. To be implemented according to ISO 4414. Typical residual risk: - Failure to follow documented procedures	
	Push buttons	**Hold-to-run control** Implemented according to IEC 60204-1. with push buttons only trained personnel who understands the hazardous movements are permitted to access these pushbuttons which cant be accidentally reached from outside protected with guard locking function Typical residual risk: - Control of hazard by person other than exposed operator - Inappropriate design of control device **Assigned Safety Function(s):** SF03 – Disabling safety functions manually and for limited time	PLr c
	Reset	**Reset button** To be implemented according to ISO 13849-1 and IEC 60204-1. Blue colored push button with indicator lights recommended. Typical residual risk: - Inappropriate modification of equipment/fastening solution - Failure to confirm safeguarding space clear of other individuals - Control of reset by person other than exposed operator	PLr d
	T001	**Interlock - Guard locking** To be implemented according to ISO 14119. Paired with Guard locking function. Typical residual risk: - Improper location of interlocking guard relative to hazard zone(s) - Inappropriate modification of equipment/fastening solution - Use of uncontrolled spare actuators - Improper parameterization of unlock signal - Failure due to mechanical wear **Assigned Safety Function(s):** SF13 – Safeguarding work areas shared by humans and machines	PLr d
	T004	**E-Stop push button** Additional estop button to be implemented according to ISO 13850 and IEC 60204-1. Typical residual risk: - Operation for purposes other than emergency	PLr d

		- Device not readily accessible - Introduction of obstruction preventing actuation **Assigned Safety Function(s):** SF99 – Emergency stop	
	T008	**Emergency off** To be implemented according to IEC 60204-1. Typical residual risk: - Failure to follow documented procedures - Operation for purposes other than emergency - Device not readily accessible	
Administrative measures			
	A001	**Instruction manual** Machine manual(s) to be updated and available in accordance to the requirements of Machinery Directive 2006/42/EC. Machine manual(s) to be available in accordance to the local/regional requirements. Typical residual risk: - Failure to follow documented instruction - Documentation not readily available to operator - Documentation not in appropriate language(s)	
	A005	**Lockout / Tagout** To be implemented as supplemental measure to ISO 14118. Paired with device Machine specific LOTO procedure with SOP mix . Typical residual risk: - Failure to follow documented procedures	
	A006	**Safe work procedures** Safe work procedures to be developed in accordance to the requirements of Use of Work Equipment Directive 2009/104/EC. Safe work procedures to be developed in accordance to the local/regional requirements. Typical residual risk: - Failure to follow documented procedures - Procedure not provided or unknown to operator	
	A010	**Measures to avoid unauthorized access** To implement measures to prevent unauthorized access or memory alteration according to the requirements stated in IEC 60204-1 and/or regional/local regulations. Typical residual risk: - Key/password accessible by unauthorized personnel	
	Crush Injury	**Awareness (safety) signs** To implement safety awareness sign that complies to ISO 7010. To implement awareness sign as according to the requirements stated in ANSI B11.19. Typical residual risk: - Failure to heed warning - Failure to maintain visibility/legibility	
	Guards	**Awareness (safety) signs** To implement safety awareness sign that complies to ISO 7010. To implement awareness sign as according to the requirements stated in ANSI B11.19. Typical residual risk: - Failure to heed warning - Failure to maintain visibility/legibility	
	Hot surface	**Awareness (safety) signs** Hot surface temperature is more than acceptable limits of human to handle by bear hands Typical residual risk: - Failure to heed warning - Failure to maintain visibility/legibility	
	I&M	**Instruction manual** Machine manual(s) to be updated and available in accordance to the requirements of Machinery Directive 2006/42/EC. Machine manual(s) to be available in accordance to the local/regional requirements. Typical residual risk: - Failure to follow documented instruction	

		- Documentation not readily available to operator - Documentation not in appropriate language(s)	
	LOTO Policy	**Lockout / Tagout** To be implemented as supplemental measure to ISO 14118. Paired with device Machine specific LOTO procedure with SOP mix . Typical residual risk: - Failure to follow documented procedures	
	PPE	**PPE - General** General PPE as recommended and specified by the machine builder and/or machine owner for noise and dust present near the machine area Typical residual risk: - Failure to properly use and maintain PPE	
	Security screws	**Tamper resistant fastener** To implement tamper resistant fixings according to the requirements in ISO 14119. Typical residual risk: - Special tool accessible by unauthorized personnel - Failure to replace tamper resistant fastener after completion of task **Assigned Safety Function(s):** SFXX – Tamper proof	
	USER PW	**Measures to avoid unauthorized access** To implement measures to prevent unauthorized access or memory alteration according to the requirements stated in IEC 60204-1 and/or regional/local regulations. Typical residual risk: - Key/password accessible by unauthorized personnel	

7 Risk assessment process

7.1 Risk assessment

The level of risk must first be determined and documented as the first step in determining safeguarding measures to reduce the risk of machine hazards. The methodology used by SICK to assess risk is in compliance with the processes described in ISO 12100 and ANSI B11 and is proven to establish a systematic approach to identify, evaluate and reduce risks for people as required by regulations for machine safety. This process must be documented as evidence of the information used to select the appropriate risk reduction measures and determine their minimum performance requirements, as appropriate.

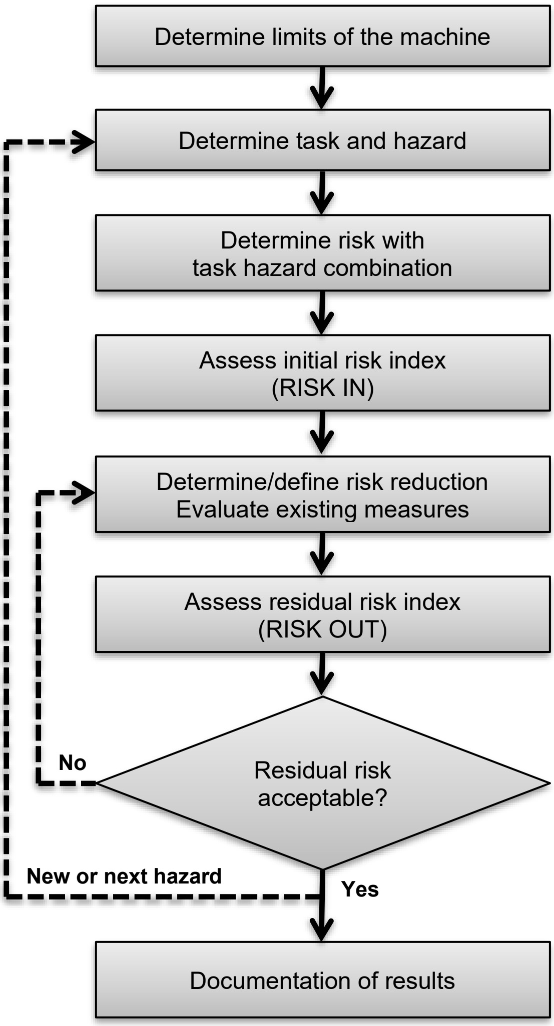


Figure 1: Overview of risk assessment process

7.2 Determination of the machine limits

7.2.1 Use Limits

Use limits include the intended use as well as the reasonably foreseeable misuse of the equipment. Aspects to be taken into account include the following:

- **a)** The different machine operating modes and the different intervention procedures for the users (including interventions required by malfunctions of the machine use),
- **b)** the use of the machinery (e.g., industrial, non-industrial and domestic) by persons identified by sex, age, dominant hand usage, or limiting physical abilities (e.g., visual or hearing impairment, size, strength) – if specific information is not available, the assessment should take into account general information about the intended user population (e.g., appropriate anthropometric data),
- **c)** the anticipated levels of training, experience, or ability of users, such as:
  - **1)** Operators
  - **2)** Maintenance personnel or technicians
  - **3)** Trainees and apprentices
  - **4)** The general public
- **d)** exposure of other individuals to the hazards associated with the machinery where it can be reasonably foreseen, including:
  - **1)** Operators working in the vicinity, e.g., operators of adjacent machinery (i.e., persons likely to have good awareness of the specific hazards)
  - **2)** Non-operator employees in the vicinity, e.g., administrative staff (i.e., persons with little awareness of specific hazards but likely to have good awareness of site safety procedures, authorized routes, etc.)
  - **3)** Non-employees in the vicinity, e.g., visitors (i.e., persons likely to have very little awareness of the machine hazards or the site safety procedures; members of the public, including children where applicable)

7.2.2 Space Limits

Aspects to be taken into account include:

- **a)** Range of movement
- **b)** Space requirements for persons to interact with the machine, e.g., during operation and maintenance
- **c)** Human interaction, e.g., “operator-machine” interface
- **d)** “Machine-power supply” interface

7.2.3 Time Limits

Aspects to be taken into account include:

- **a)** The “life limit” of the machinery and/or of some of its components (e.g., tooling, parts that can wear, electromechanical components), taking into account its intended use and reasonably foreseeable misuse
- **b)** Recommended service intervals

7.3 Determination of tasks and hazards

Following the determination of the limits of the machinery, the essential step in any machine risk assessment is the systematic identification of reasonably foreseeable hazards, hazardous situations and/or hazardous events during all phases of the machine life cycle, i.e.:

a) Transport, assembly and installation
b) Commissioning
c) Use
d) De-commissioning, dismantling and disposal

It is assumed that, when present on machinery, a hazard will sooner or later lead to harm if measures are not taken to eliminate or provide protective measures. Only when hazards have been identified can steps be taken to eliminate them or reduce risks. To accomplish this hazard identification, it is necessary to identify the operations to be performed by the machinery and the tasks to be performed by persons who interact with it, taking into account the different parts, mechanisms or functions of the machine, the materials to be processed, if any, and the environment in which the machine can be used. Task identification should consider all those tasks associated with all the phases of the machine life cycle listed above as considered in the scope of work. Task identification should also take into account, but not be limited to, the following task categories:

a) Setting
b) Testing
c) Teaching/programming
d) Process/tool changeover
e) Start-up
f) All modes of operation
g) Feeding machine
h) Removal of product from machine
i) Stopping the machine
j) Stopping the machine in an emergency
k) Recovery of operation from jam
l) Re-start after unscheduled stop
m) Faultfinding/troubleshooting (operator intervention)
n) Cleaning and housekeeping
o) Preventive maintenance
p) Corrective maintenance

The aim is to identify all reasonably foreseeable hazards, hazardous situations or hazardous events.

7.4 Risk Estimation

The risk assessment methodology utilized by SICK is designed to easily estimate and then distinguish between high and low risk situations. This process is aimed at producing consistently reproducible and easily understandable results. In order to provide either rough or detailed estimations of the risk parameters, consideration of secondary and sometimes tertiary matrices can be utilized to adapt the process to the requirements and resources of each application. For each risk a risk estimation will be performed using the Scalable Risk Analysis and Evaluation Method [SCRAM] based on a primary evaluation of the risk parameters.

Severity	Exposure	Avoidance	Occurrence				PLr
			O1 – O3	O1	O2	O3	
S1	E0	÷	< 1				a
	E1 – E3	A1, A2		< 1	< 1	< 1	
S2	E0	÷	≤ 1				
	E1	A1		< 1	< 1	1	b
		A2		< 1	1	1	
	E2	A1		1	2	2	c
		A2		1	2	2	
	E3	A1		2	3	3	
		A2		2	3	3	
S3	E0	÷	1				
	E1	A1		3	4	4	d
		A2		3	4	4	
	E2	A1		4	5	5	
		A2		5	5	5	
	E3	A1		5	6	6	
		A2		6	6	6	
S4	E0	÷	1				
	E1	A1		6	7	7	
		A2		7	7	7	
	E2	A1		7	8	8	e
		A2		8	8	8	
	E3	A1		8	9	9	
		A2		9	10	10	
			Risk Index				

Figure 2: Scalable Risk Analysis and Evaluation Method [SCRAM]

Parameter		**Severity of harm S** (simplified estimation)
S1	NEGLIGIBLE	None or negligible (trivial) injury (e.g., small bruises or superficial cuts) which either do not require any treatment or only treatment that is limited to simple and normally available first aid methods and equipment.
S2	SLIGHT	Injuries which can be treated with normally available first aid equipment but require the help of medically trained personnel. *- OR –* The injury (medical condition) will be reversed within three months without treatment, but under monitoring of a medical practitioner.
S3	SERIOUS	Injuries which require treatment by a medical practitioner but do not lead to permanent impairment. *- OR –* Injuries which lead to the loss or permanent damage of parts of the human body (but not total loss) with reversible medical condition.
S4	SEVERE	Injuries which lead to the death of one or more persons. *- OR –* Injuries which require treatment by a medical practitioner in a hospital and may lead to a permanent impairment or loss of parts of the body, limbs, or senses/abilities.

Parameter		**Exposure to hazard E** (simplified estimation)
E0	PREVENTED	Foreseeable exposure or access to the hazard(s) is: eliminated / controlled / limited by inherently safe design measures, *- OR -* exposure is prevented by mechanical and/or control-related safeguarding equipment which is selected and implemented as appropriate for the application. The implemented functional safety performance of the related SRP/CS must meet or exceed the required functional safety performance (PL ≥ PLr). E0 not available for initial estimation of risk
E1	LOW	LOW (E1) can always be assumed if: the exposure frequency is once per work shift, *- AND -* the exposure duration is less than 3 minutes.
E2	MEDIUM	Use level 2 for determination of exposure to hazard. For details see: Whitepaper "Risk Assessment and Risk Reduction for Machinery Part 3".
E3	HIGH	HIGH (E3) can always be assumed, if: access to the hazard zone is required, *- AND -* the exposure frequency is at least twice per work shift, *- AND -* each time the exposure duration is equal to or higher than 1 minute.

Parameter		**Possibility of avoidance A** (simplified estimation)
A1	AVOIDABLE	There are certain conditions that allow the avoidance of harm (such as skilled workers, slow movements, infrequent intervention, low-complexity processes, no sudden or unexpected movements with high acceleration).
A2	NOT AVOIDABLE	The avoidance is nearly impossible due to the lack of indication or awareness of the hazardous situation (such as fast hazardous events, insufficient surrounding space for evasion, high complexity processes, and/or the effect of routine on hazard awareness).

Parameter		**Probability of occurrence O** (simplified estimation)
O1	LOW	LOW (O1) can always be assumed if: the system is not prone to trouble, - AND - the hazard frequency is less than 33% of the presence time.
O2	MEDIUM	Use level 2 for determination of probability of occurrence. For details see: Whitepaper "Risk Assessment and Risk Reduction for Machinery Part 3".
O3	HIGH	HIGH (O3) can always be assumed if: the system is prone to trouble, - AND - the hazard frequency is equal to or higher than 66% of the presence time.

7.5 Risk Evaluation

The initial risk estimation [RISK IN] describes the risk without considering any measures in place. This risk index describes the level of the existing risk on a scale from <1 (low risk) to 10 (very critical risk). Following risk estimation, a risk evaluation must be carried out to determine if risk reduction is required. If risk reduction is determined to be necessary, appropriate protective measures shall be selected and applied, and the risk assessment process repeated. As part of this iterative process, the designer must check whether additional hazards are introduced, or other risks increased when new risk reduction measures are applied. If additional hazards do occur, they shall be added to the list of identified hazards and appropriate protective measures will be required to address them.

7.6 Selection of Risk Reduction Measures

When the evaluation process determines that additional risk reduction measures are necessary, the selection and application of protective measures should be a decision made according to the risk(s) associated with each application.
When selecting risk reduction measures, it is important to consider that different measures have different effects on the various risk factors. With this in mind, it is always prudent to apply more than one protective measure in order to have the greatest cumulative effect on the reduction of risk. Furthermore, some protective measures are more susceptible to failures or error/misuse than others.

7.7 Evaluate Residual Risk

Following application of one or more risk reduction measures, the residual risk must again be assessed, this time accounting for the protective measures that are already in place as well as any new measures being added to the overall solution. In this report, the estimated residual risk is identified as RISK OUT and is based on the **assumption** that all recommended protective measures have been **implemented and validated according to applicable standards** and will be maintained over time through periodic inspection and maintenance.

The use of guards or safety functions very often results in people no longer being exposed to hazards or having no access. In this case, the parameter [E0] in the exposure group is used to estimate the residual risk. A hazard at which the probability of occurrence is prevented (E0) and additional administrative measures provided by the designer have been implemented by the user is rated with a residual risk index (RISK OUT) of AR (As low as reasonably practicable).

7.8 Achieving Acceptable (Tolerable) Residual Risk

It is important to understand that the intent of the risk assessment process is to identify risks which are not acceptable, apply protective measures based on the level of inherent risk present, and ultimately to reduce the risk to a level deemed acceptable or tolerable by both the organization as well as the affected individuals. While the aim is to achieve residual risk levels that are low or negligible, it is important to point out that the attainable level of residual risk cannot always achieve such a low degree. Instead, acceptable residual risk is often equated to the concept of ‘As Low As Reasonably Practicable’ (ALARP), which is a common best practice to judge the balance of risk and societal benefit.

> The subjective nature of determining if acceptable risk has been achieved precludes SICK from stating whether the residual risk achievable is tolerable to any organization. As a partner with specialized experience and technical expertise, SICK can provide guidance and recommendations to your organization. In the end, however, acceptance (or rejection) of residual risk is the sole responsibility of the organization choosing to place or maintain industrial equipment in the workplace.

8 Information data protection

(Pursuant to Article 13 GDPR)

Subject	Description
Responsible body	SSU – place of jurisdiction Sick India Pvt. ltd. 215, Western Edge II, Western Express Highway, 400066 Mumbai
Purpose of data processing	The Machine Safeguarding Evaluation service supports the purpose to gain knowledge about the safety status of one or more machines in order to recommend appropriate protective measures in the following step
Categories of data	▪ Customer address ▪ Photos/Sketches/Graphics ▪ Writing down interviews
Legal basis	GDPR Art. 6 (1) f) – corporate interests
Corporate interests	Sick has the following interest: ▪ Accurate / verifiable MSE ▪ Storage for proof reasons (§199 Abs. 2 BGB) ▪ Interest of the individual worthy of protection – No detailed report (anonymized)
Consequences of non-delivery of data	The provision of personal data is not required by law or contract. If the required data is refused by the customer, the quality of the evaluation decreases or the evaluation becomes impossible.
Data origin	30 years after preparation of the report
Data origin	SICK only processes data which have been provided to SICK directly or that are originated from the findings of the Machine Safeguarding Evaluation service.
Recipient of the data	SICK AG Erwin-Sick-Straße 1 79183 Waldkirch Germany
Transfers to third countries	No third countries
Logic of the automated decision	No automated decisions are made

You have the right
• for information about personal data concerning you;
• to correct personal data concerning you;
• to delete or block personal data concerning you;
• to object to the processing of personal data concerning you;
• the transferability of personal data concerning you;
• to revoke your consent with effect for the future, provided that any processing of personal data concerning you is based on your consent; and
• complain to a data protection supervisory authority if you believe that the processing of personal data concerning you violates data protection law.

9 Abbreviations

Information of used acronyms / customer definitions	
AOPD	Active opto-electronic protective device
AOPDDR	Active opto-electronic protective device responsive to diffuse reflection
ESPE	Electro sensitive protective device
PL	Performance level
PLr	Performance level required
PPE	Personal protective equipment

10 Risk reduction measures symbols

	Discard measure
	Maintain measure
	Manipulation apparent
	Modify measure
	New recommended measure
	Replace existing device with similar device
	Replace existing measure with other measure

11 Signature

SICK Safety Specialist

Name	
Title	
Date / Signature	

SICK Safety Specialist

Name	
Title	
Date / Signature	

Customer Acknowledgement

Name	
Title	
Date / Signature	

Customer Acknowledgement

Name	
Title	
Date / Signature	

SICK AT A GLANCE

SICK is a leading manufacturer of intelligent sensors and sensor solutions for industrial applications. With more than 11,000 employees and over 50 subsidiaries and equity investments as well as numerous agencies
worldwide, we are always close to our customers. A unique range of products and services creates the perfect basis for controlling processes securely and efficiently, protecting individuals from accidents and preventing damage to the environment.

We have extensive experience in various industries and understand their processes and requirements. With intelligent sensors, we can deliver exactly what our customers need. In application centers in Europe, Asia and North America, system solutions are tested and optimized in accordance with customer specifications. All this makes us a reliable supplier and development partner.

Comprehensive services round out our offering: SICK LifeTime Services provide support throughout the machine life cycle and ensure safety and productivity.

For us, that is “Sensor Intelligence.”

Worldwide presence

Australia, Austria, Belgium, Brazil, Canada, Chile, China, Czech Republic, Denmark, Finland, France, Germany, Great Britain, Hungary, India, Israel, Italy, Japan, Malaysia, Mexico, Netherlands, New Zealand, Norway, Poland, Romania, Russia, Singapore, Slovakia, Slovenia, South Africa, South Korea, Spain, Sweden, Switzerland, Taiwan, Thailand, Turkey, United Arab Emirates, USA, Vietnam.

Detailed addresses and further locations: → **www.sick.com**